EE-102: Electric Circuit Analysis

Lecture # 01

About Me

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★ *Email me if you have any query….. I will not respond to any query on whatsapp.*

Policies

- The Lecture will always start on time, otherwise the changed schedule/time table will be communicated with you in advance.
- The quiz once taken will not be re-taken. There will be no makeup quiz.
- Assignments can be submitted before or on mentioned time.
  - If submitted after 24 hours of the due time, half marks shall be granted.
  - After that zero marks will be given.

Honesty is the best Policy!

- **Penalty for Cheating Cases:**
  - First Offense: Zero in assignment for all involved.
  - Further Offenses: Zero in assignment and -2 from internals/pre-finals theory marks for all involved.

Organization of Lecture

- 1 Introduction to Circuits and Networks
  - 1.1 Basic Phenomena
  - 1.2 Ideal Elements
  - 1.3 Electric Circuits
  - 1.4 Units
  - 1.5 Definitions of Various Terms
  - 1.6 Steady State Analysis and Transient Analysis
  - 1.7 Assumptions in Circuit Theory
  - 1.8 Solved Examples

1.1 Basic Phenomena

- The energy associated with flow of electrons is called **electrical energy**. The flow of electrons is called **current**. The current can flow from one point to another point of an element only if there is a **potential difference** between these two points. The potential difference is called **voltage**.
- When electric current is passed through a device or element, three phenomenas have been observed. The three phenomenas are,

  i. opposition to flow of current (Resistor).

  ii. opposition to change in current or flux (Inductor).

  iii. opposition to change in voltage or charge (Capacitor).

1.1 Basic Phenomena (continued…)

- The various effects of current like heating, arcing, induction, charging, etc., are due to the phenomena discussed in previous slide. Therefore, three fundamental elements have been proposed which exhibit only one of the phenomena when considered as an ideal element (of course, there is no ideal element in nature). These elements are resistor, inductor and capacitor.

1.2 Ideal Elements

- The **ideal resistor** offers opposition only to the flow of current. The property of opposition to the flow of current is called **resistance** and it is denoted by **R**.
- The **ideal inductor** offers opposition only to change in current (or flux). The property of opposition to change in current is called **inductance** and it is denoted by **L**.
- The **ideal capacitor** offers opposition only to change in voltage (or charge). The property of opposition to change in voltage is called **capacitance** and it is denoted by **C**.

1.3 Electric Circuits

- The behaviour of a device/element to electric current can be best understood if it is modelled using the fundamental elements R, L and C. For example, an incandescent lamp and a water heater can be modelled as ideal resistance. Transformers and motors can be modelled using resistance and inductance.
- Practically, an **electric circuit** is a model of a device operated by electrical energy. The various concepts and methods used for analysing a circuit is called **circuit theory**. A typical circuit consists of source(s) of electrical energy and ideal elements R, L and C. The practical energy sources are batteries, generators (or alternators), rectifiers, transistors, op-amps, etc. The various elements of electric circuits are shown in Figs 1 and 2 next slides.

1.3 Electric Circuits (continued…)

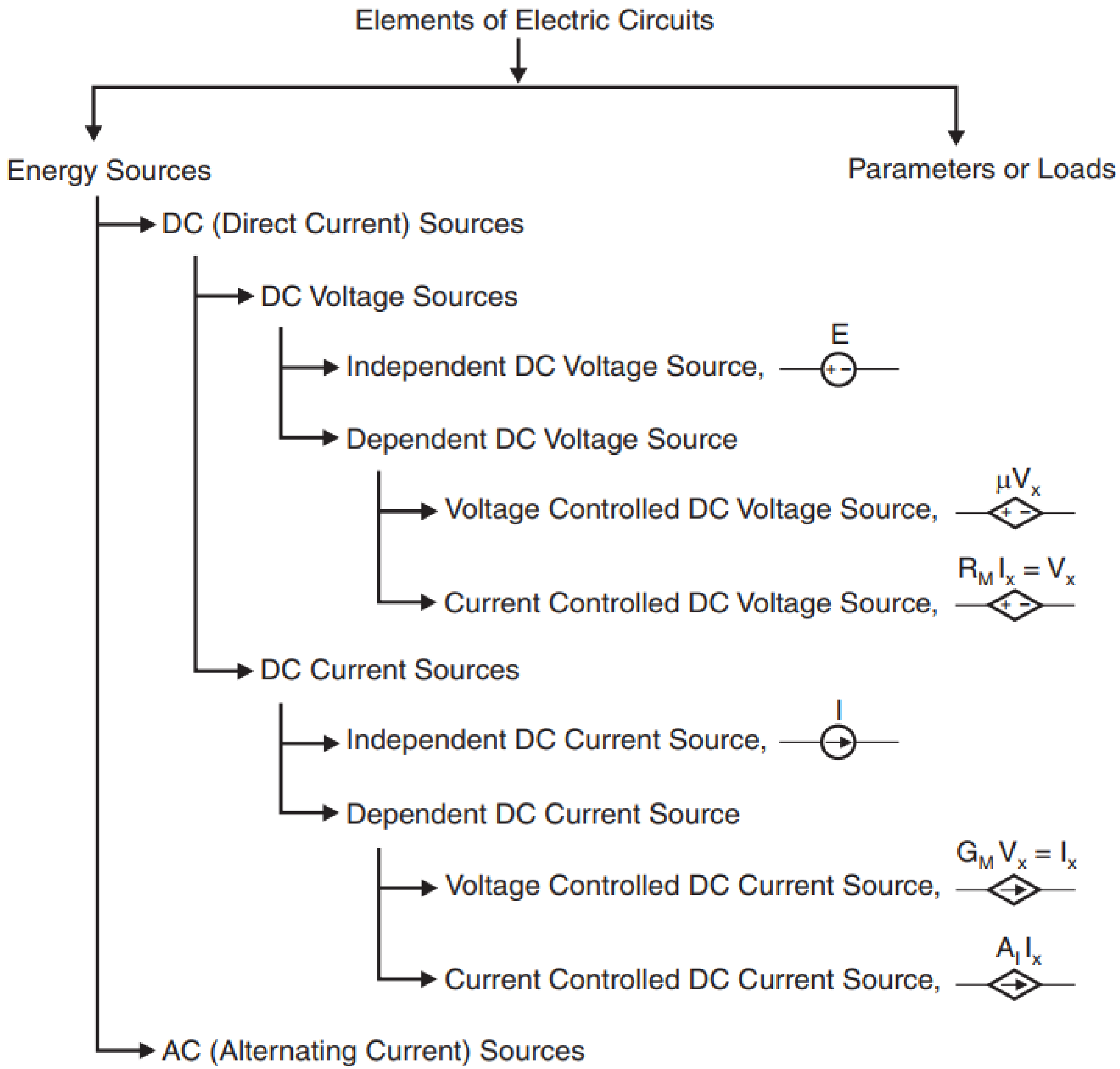


Figure 1: Elements of Electric circuits – Energy source

1.3 Electric Circuits (continued…)

- AC (Alternating Current) Sources
  - AC Voltage Sources
    - Independent AC Voltage Source, $\bar{E} = E\angle\theta^{\circ}\ V$
    - Dependent AC Voltage Source
      - Voltage Controlled AC Voltage Source, $\mu\bar{V}_x$
      - Current Controlled AC Voltage Source, $R_M\,\bar{I}_x = \bar{V}_x$
  - AC Current Sources
    - Independent AC Current Source, $\bar{I} = I\angle\theta^{\circ}\ A$
    - Dependent AC Current Source
      - Voltage Controlled AC Current Source, $G_M\,\bar{V}_x = \bar{I}_x$
      - Current Controlled AC Current Source, $A_I\,\bar{I}_x$

Figure 1: Elements of Electric circuits – Energy source (continued)

1.3 Electric Circuits (continued...)

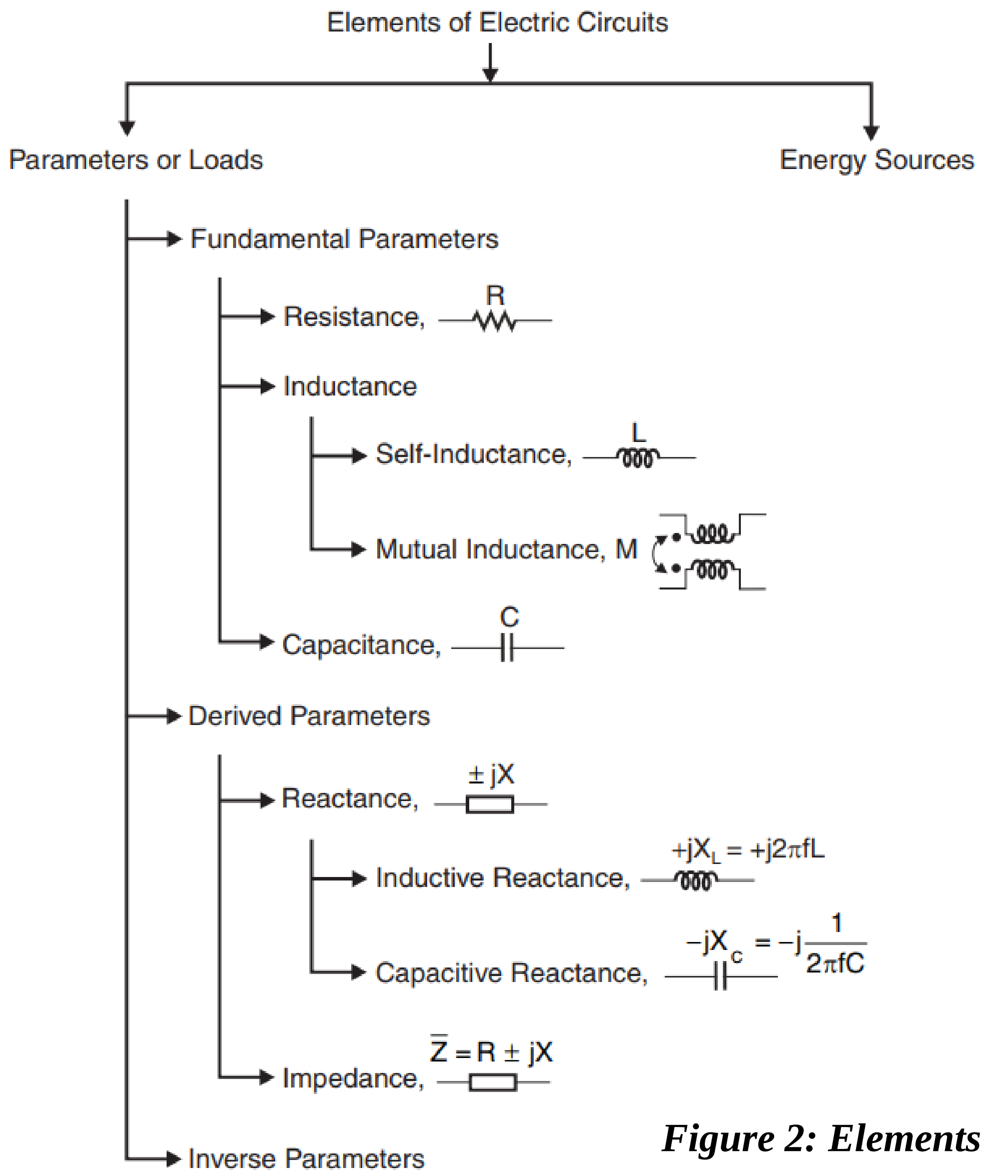


Figure 2: Elements of Electric circuits – Parameters or loads

1.3 Electric Circuits (continued...)

- Inverse Parameters
  - Conductance, $G = \dfrac{1}{R}$
  - Susceptance, $\mp jB = \dfrac{1}{\pm jX}$
    - Inductive Susceptance, $-jB_L = -j\dfrac{1}{2\pi fL}$
    - Capacitive Susceptance, $+jB_C = +j2\pi fC$
  - Admittance, $\overline{Y} = G \mp jB$

$$\overline{Y} = \frac{1}{\overline{Z}} = \frac{1}{R \pm jX} = G \mp jB$$

Figure 2: Elements of Electric circuits – Parameters or loads (continued)

1.3 Electric Circuits (continued…)

- Elements which generate or amplify energy are called **active elements**. Therefore, energy sources are active elements. Elements which dissipate or store energy are called **passive elements**. ***Resistance*** *dissipates energy in the form of* ***heat***, ***inductance*** *stores energy in a* ***magnetic field***, and ***capacitance*** *stores energy in an* ***electric field***. Therefore, resistance, inductance and capacitance are passive elements. If there is no active element in a circuit then the circuit is called a **passive circuit** or **network**.
- Sources can be classified into independent and dependent sources. Batteries, generator sand rectifiers are independent sources, which can directly generate electrical energy. Transistors and op-amps are dependent sources whose output energy depends on another independent source.

1.3 Electric Circuits (continued...)

- Practically, the sources of electrical energy used to supply electrical energy to various devices like lamps, fans, motors, etc., are called **loads**. The rate at which electrical energy is supplied is called **power**. Power, in turn, is the product of voltage and current.
- Circuit analysis relies on the concept of **law of conservation of energy**, which states that energy can neither be created nor destroyed but can be converted from one form to other. Therefore, the total energy/power in a circuit is zero.

1.4 Units

- The **SI units** and their symbols for various quantities encountered in circuit theory are presented in Table 1 below:

Table 1: Units and Symbols

Quantity	Symbol for quantity	Unit	Unit symbol	Equivalent unit	Equivalent unit symbol
Charge	q, Q	Coulomb	C	-	-
Current	i, I	Ampere	A	Coulomb/second	C/s
Flux linkages	ψ	Weber-turn	Wb	-	-
Magnetic flux	ϕ	Weber	Wb	-	-
Energy	w, W	Joule	J	Newton-meter	N-m
Voltage	v, V	Volt	V	Joule/Coulomb	J/C
Power	p, P	Watt	W	Joule/second	J/s
Capacitance	C	Farad	F	Coulomb/Volt	C/V
Inductance	L, M	Henry	H	Weber/Ampere	Wb/A
Resistance	R	Ohm	Ω	Volt/Ampere	V/A
Conductance	G	Siemens	S	Ampere/Volt or mho	A/V *or* ℧

1.4 Units (continued...)

Table 1: (continued…)

Quantity	Symbol for quantity	Unit	Unit symbol	Equivalent unit	Equivalent unit symbol
Time	t	Second	s	-	-
Frequency	f	Hertz	Hz	cycles/second	-
Angular frequency	ω	Radians/second	rad/s	-	-
Magnetic flux density	-	Tesla	T	Weber/ meter square	Wb/m^2
Temperature	-	Kelvin	^{o}K	-	-

1.4 Units

Table 2: Multiple and Submultiple used for Units

Multiplying factor	Prefix	Symbol
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10^{1}	deca	da

Multiples used for Units

Multiplying factor	Prefix	Symbol
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

Submultiples used for Units

1.5 Definition of Various Terms

- The definitions of various terms that are associated with electrical energy like energy, power, current, voltage, etc., are presented in this section.
- **Energy** is defined as the capacity to do work. Energy may exist in many forms, such as electrical, mechanical, thermal, light, chemical, etc. It is measured in joules, which is denoted by ***J*** (or the unit of energy is joules).
- In electrical engineering, one **joule** is defined as the energy required to transfer a power of one watt in one second to a load (or Energy = Power × Time). Therefore, $1J = 1W\text{-}s$.

1.5 Definition of Various Terms (continued…)

- **Power** is the rate at which work is done (or it is the rate of energy transfer). The unit of power is watt and denoted by ***W***. If energy is transferred at the rate of one joule per second then one **watt** of power is generated.
- An average value of power can be expressed as,

$$\text{Power, } P = \frac{\text{Energy}}{\text{Time}} = \frac{W}{t}$$

A time varying power can be expressed as,

$$\text{Instantaneous power, } p = \frac{dw}{dt}$$

$$\text{Also, } p = \frac{dw}{dt} = \frac{dw}{dq} \times \frac{dq}{dt} = v\, i$$

where, dq = Change in charge in a time of dt

dt = Time required to produce a change in charge dq

1.5 Definition of Various Terms (continued…)

- **Charge** is the characteristic property of elementary particles of matter. The elementary particles are electrons, protons and neutrons. There are basically two types of charges in nature: positive charge and negative charge. The charge of an electron is called **negative charge**. The charge of a proton is called **positive charge**. Normally, a particle is neutral because it has equal number of electrons and protons. The particle is called charged if some electrons are either added or removed from it. If electrons are added then the particle is called negatively charged. If electrons are removed then the particle is called positively charged. The unit used for measurement of charge is **coulomb**.
- **One coulomb** of charge is the total charge associated with 6.242×10^{18} electrons.

1.5 Definition of Various Terms (continued…)

- **Current** is defined as the rate of flow of electrons. It is measured in amperes.
- If 6.242 x 10^{18} electrons (1 coulomb of charge) pass through the imaginary plane in Fig. 3 (below) in 1 second, the flow of charge, or current, is said to be 1 **ampere** (A).

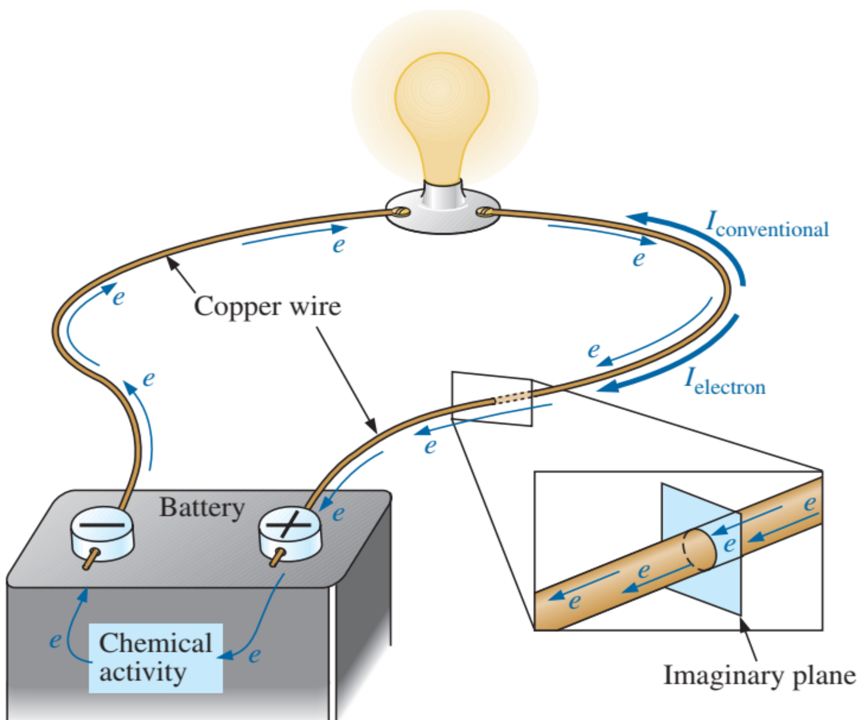


Fig.3: Basic Electric Circuit

1.5 Definition of Various Terms (continued…)

A steady current can be expressed as,

$$\text{Current, } I = \frac{\text{Charge}}{\text{Time}} = \frac{Q}{t}$$

A time varying current can be expressed as,

$$\text{Instantaneous current, } i = \frac{dq}{dt}$$

where, Q = Charge flowing at a constant rate

t = Time

dq = Change in charge in a time of dt

dt = Time required to produce a change in charge dq

1.5 Definition of Various Terms (continued…)

- Every charge will have potential energy. The difference in potential energy between the charges is called **potential difference**. In electrical terminology, the potential difference is called **voltage**. *Potential difference indicates the amount of work done to move a charge from one place to another*. Voltage is expressed in volt. One **volt** is the potential difference between two points, when one joule of energy is utilised in transferring one coulomb of charge from one point to the other. A steady voltage can be expressed as,

$$\text{Voltage, } V = \frac{\text{Energy}}{\text{Charge}} = \frac{W}{Q}$$

A time varying voltage can be expressed as,

$$\text{Instantaneous voltage, } v = \frac{dw}{dq}$$

$$\therefore \text{ Voltage, } V = \frac{\text{Power}}{\text{Current}} = \frac{P}{I} \longrightarrow$$

Recall !

P=VI

1.5 Definition of Various Terms (continued…)

- An **electrical signal** is a voltage or current which conveys information, usually it means a voltage. The term can be used for any voltage or current in a circuit.
- The voltage-time graph given below shows various properties of an electrical signal. In addition to the properties labelled on the graph, there is frequency which is the number of cycles per second.

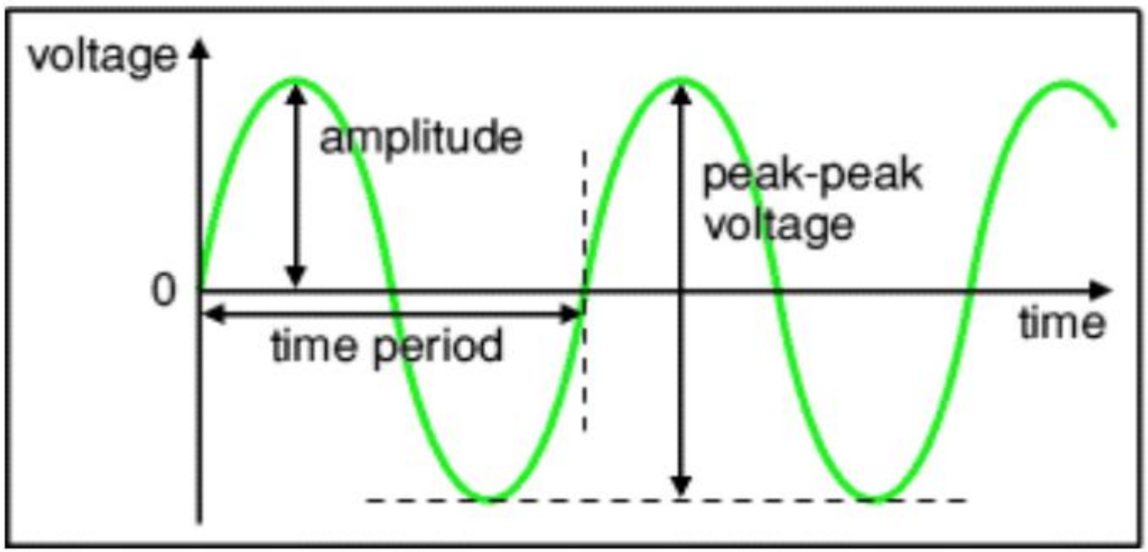


Fig.4: Electrical Signal- Voltage-time graph

1.5 Definition of Various Terms (continued...)

- Figure 4 in previous slide shows a **sine wave** but these properties apply to any signal with a constant shape.
  - **Amplitude** is the maximum voltage reached by the signal. It is measured in **volts,V**.
  - **Peak** voltage is another name for amplitude.
  - **Peak-peak voltage** is twice the peak voltage (amplitude).
  - **Time period** is the time taken for the signal to complete one cycle.

  It is measured in **seconds (s)**, but time periods tend to be short so **milliseconds (ms)** and **microseconds (µs)** are often used. 1ms = 0.001s and 1µs = 0.000001s.
  - **Frequency** is the number of cycles per second.

  It is measured in **hertz (Hz)**, but frequencies tend to be high so **kilohertz (kHz)** and **megahertz (MHz)** are often used. 1kHz = 1000Hz and 1MHz = 1000000Hz.

$$\text{frequency} = \frac{1}{\text{time period}} \quad \text{and} \quad \text{time period} = \frac{1}{\text{frequency}}$$

1.6 Steady State Analysis and Transient Analysis

- Circuit analysis can be classified into steady state analysis and transient analysis. The analysis of circuits during switching conditions is called **transient analysis**. During switching conditions, the current and voltage change from one value to the other. *In purely resistive circuits this may not be a problem because the resistance will allow sudden change in voltage and current*.
- *In inductive circuits,* ***the current cannot change instantaneously***. *In capacitive circuits,* ***the voltage cannot change instantaneously***. Hence, when the circuit is switched from one state to the other, the voltage and current cannot attain a steady value instantaneously in inductive or capacitive circuits. Therefore, during switching conditions there will be a small period during which the current and voltage will change from an initial value to a final steady value. *The time from the instant of switching to the attainment of steady value is called* ***transient period***. Physically, the transient can be realised in switching of tube lights, fans, motors, etc.

1.6 Steady State Analysis and Transient Analysis (continued…)

- In certain circuits, the transient period is negligible and we may be interested only in steady value of the response. Therefore, steady state analysis is sufficient. *The analysis of circuits under steady state (i.e., by neglecting the transient period) is called* ***steady state analysis***.
- In certain circuits the transient period is critical and we may require the response of the circuit during the transient period. Some practical examples where transient analysis is vital are starters, circuit breakers, relays, etc. Transient analysis of circuits will be discussed later in this course.

1.7 Assumptions in Circuit Theory

- In circuit analysis the elements of the circuit are assumed to be linear, bilateral and lumped elements.
- In **linear elements**, the voltage-current characteristics are linear and the circuit consisting of linear elements is called **linear circuit** or **network**. The resistor, inductor and capacitor are linear elements. Some elements exhibit non-linear characteristics. For example, diodes and transistors have non-linear voltage-current characteristics. For analysis purpose, the non-linear characteristics can be linearised over a certain range of operation.
- In a **bilateral element**, the relationship between voltage and current will be the same for two possible directions of current through the element. A resistor is an example of a bilateral element On the other hand, a **unilateral element** will have different voltage-current characteristics for the two possible directions of current through the element. The diode is an example of a unilateral element.

1.7 Assumptions in Circuit Theory (continued…)

- In practical devices like transmission lines, windings of motors, coils, etc., the parameters(R, L and C) are distributed in nature. But for analysis purpose we assume that the parameters are lumped (i.e., concentrated at one place). This approximation is valid only for low frequency operations and it is not valid in the microwave frequency range.

1.8 Solved Examples

Q1:

Find the voltage between two points if 60 J of energy are required to move a charge of 20 C between the two points.

Solution: Eq. (2): $V = \frac{W}{Q} = \frac{60\ \text{J}}{20\ \text{C}} = \mathbf{3\ V}$

Q2:

Determine the energy expended moving a charge of 50 μC between two points if the voltage between the points is 6 V.

Solution: Eq. (3):

$$W = QV = (50 \times 10^{-6}\ \text{C})(6\ \text{V}) = 300 \times 10^{-6}\ \text{J} = \mathbf{300\ \mu J}$$

1.8 Solved Examples (continued...)

Q3: The charge flowing through the imaginary surface in figure 3 (given in slide 22) is 0.16 C every 64 ms. Determine the current in amperes.

Solution: Eq. (5):

$$I = \frac{Q}{t} = \frac{0.16\ \text{C}}{64 \times 10^{-3}\text{s}} = \frac{160 \times 10^{-3}\ \text{C}}{64 \times 10^{-3}\ \text{s}} = \mathbf{2.50\,A}$$

Q4: Determine how long it will take 4×10^{16} electrons to pass through the imaginary surface in figure 3(given in slide no 22) if the current is 5 mA.

Solution: Determine the charge in coulombs:

$$4 \times 10^{16}\ \cancel{\text{electrons}}\left(\frac{1\ \text{C}}{6.242 \times 10^{18}\ \cancel{\text{electrons}}}\right) = 0.641 \times 10^{-2}\ \text{C}$$

$$= 6.41\ \text{mC}$$

$$t = \frac{Q}{I} = \frac{6.41 \times 10^{-3}\ \text{C}}{5 \times 10^{-3}\text{A}} = \mathbf{1.28\ s}$$

References: Textbooks

1. *Robert L.Boylestad. (2014). Introductory Circuit Analysis. 12th Edition. Pearson Education Limited.*
2. *Allan H.Robbins, Wilhelm C.Miller. (2012). Circuit Analysis Theory and Practice. 5th Edition. Cengage Learning.*
3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6th Edition. McGraw Hill Education.*

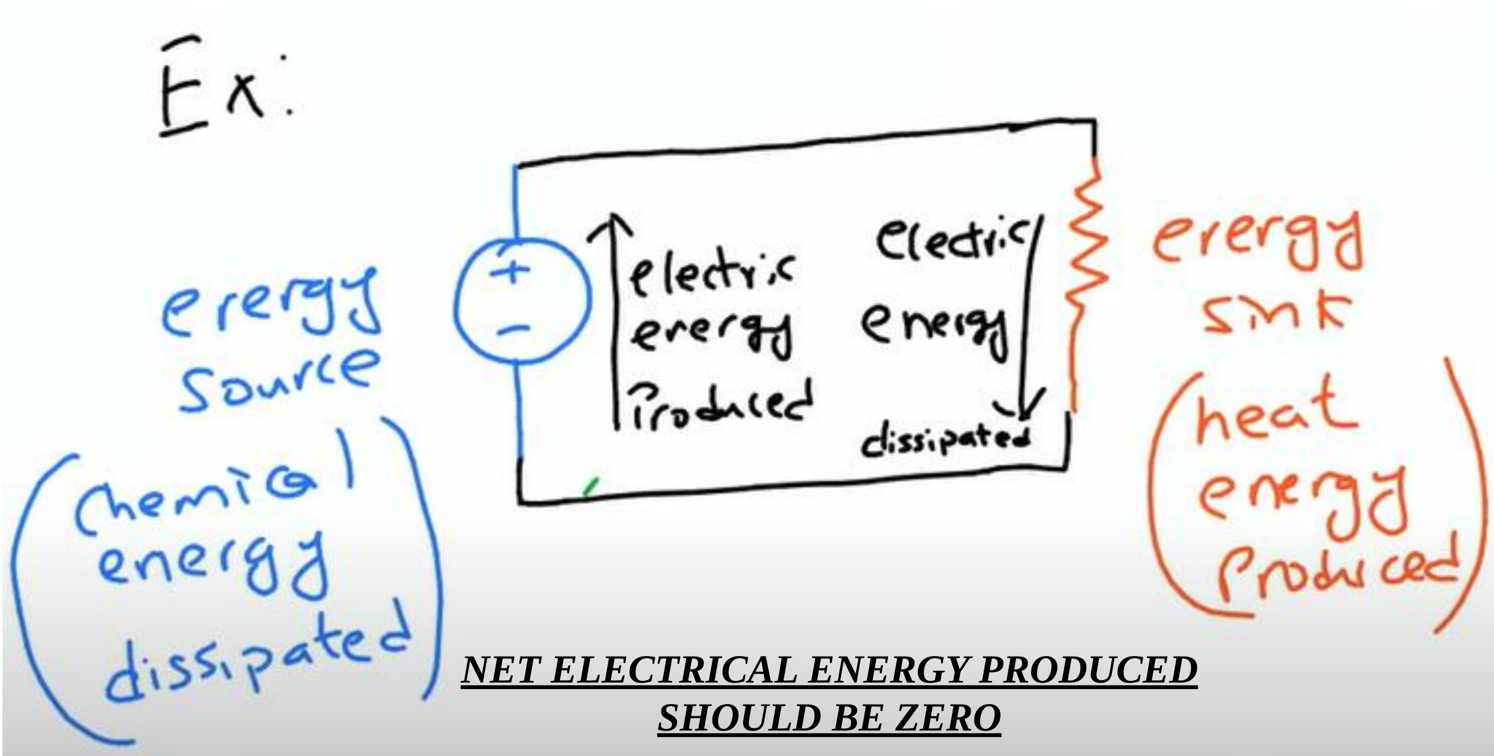
Appendix

Electric Field Versus Magnetic Field

Difference Between Electric Field vs Magnetic Field	
Electric Field	**Magnetic Field**
It creates an electric charge in surrounding	Creates an electric charge around moving magnets
Measured as newton per coulomb, volt per meter	Measured as gauss or tesla
Proportional for the electric charge	Proportional to the speed of electric charge
Are perpendicular to the magnetic field	Are perpendicular to the electric field
An electric field is measured using an electrometer	The magnetic field is measured using the magnetometer

Refer to Slide 14

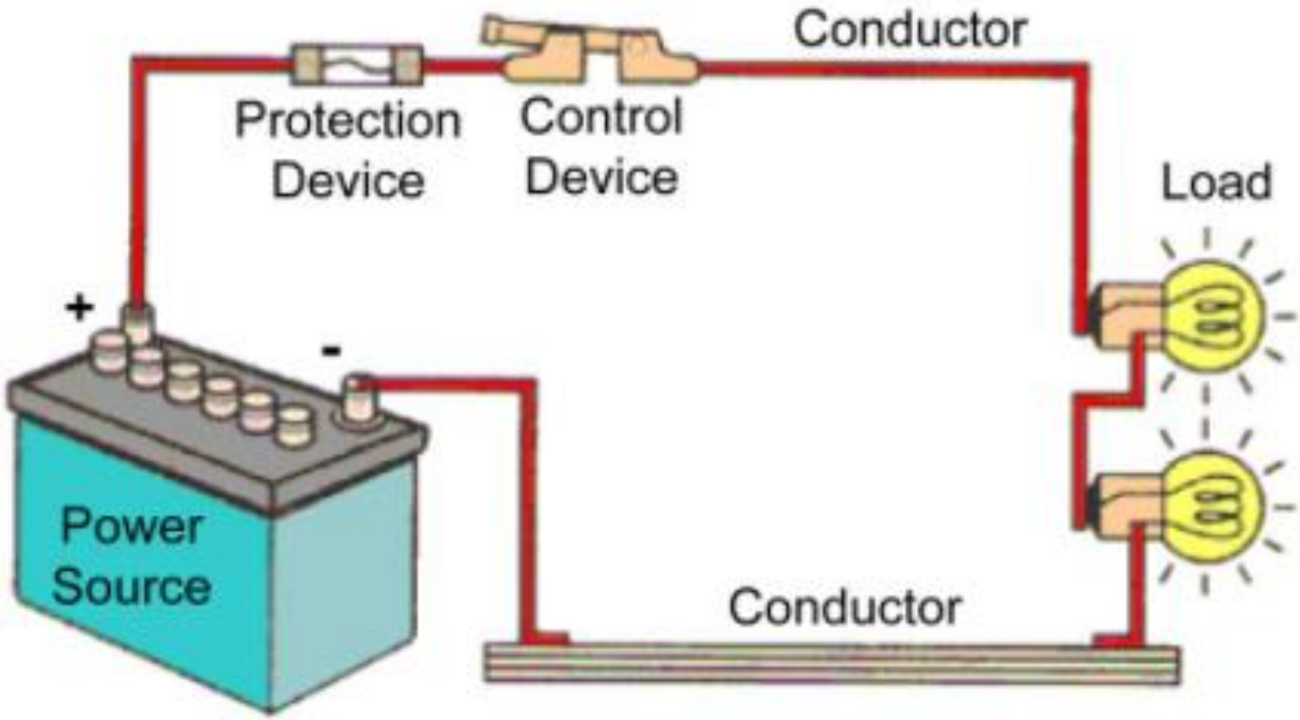
Power Balance (Law of conversation of Energy)



NET ELECTRICAL ENERGY PRODUCED SHOULD BE ZERO

Refer to Slide 15

Concept of Load(s)



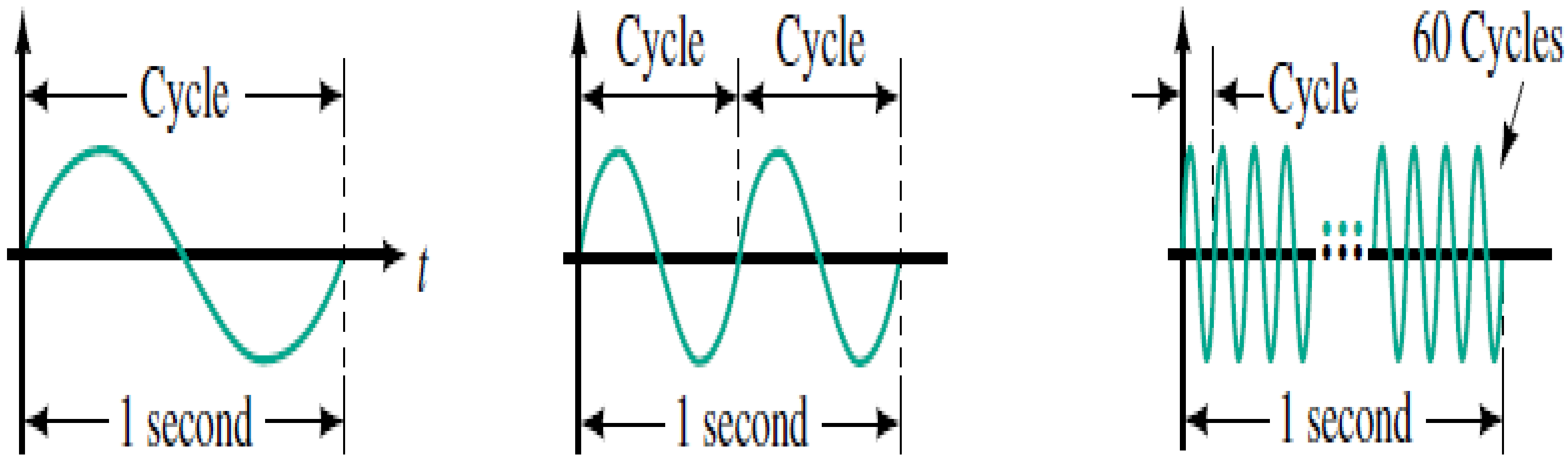
Refer to Slide 15

Frequency

- Number of cycles per second of a waveform
  - Frequency
  - Denoted by f
- Unit of frequency is hertz (Hz)
- 1 Hz = 1 cycle per second

Refer to Slide 25 and 26

Frequency Continue...



(a) 1 cycle per second = 1 Hz (b) 2 cycles per second = 2 Hz (c) 60 cycles per second = 60 Hz

Refer to Slide 25 and 26

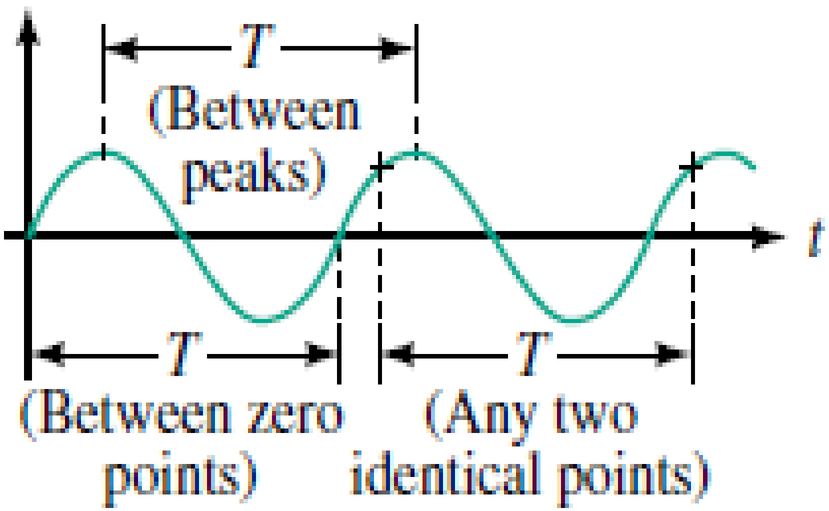
Period

- Period of a waveform
  - Duration of one cycle.
- Time is measured in seconds
- The period is the reciprocal of frequency
  - $T = 1/f\ (s)$

Refer to Slide 25 and 26

Period Continue…

- The period of a waveform can be measured between any two corresponding or identical points.
- Often it is measured between zero points.



Refer to Slide 25 and 26

Induced Voltage and Induction

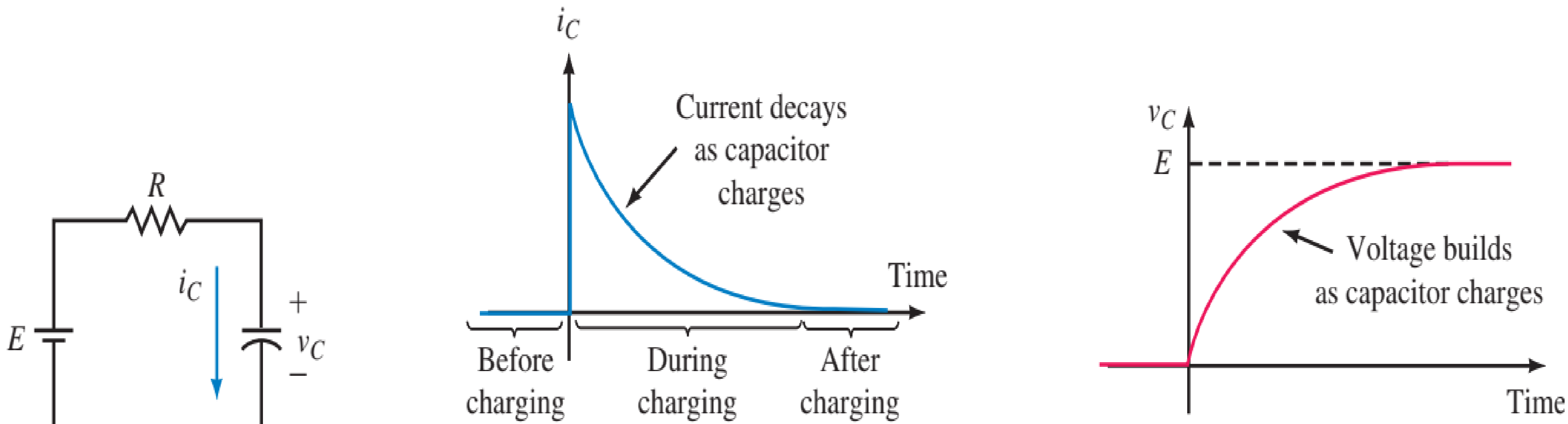
(a) Current cannot jump from one value to another like this

(b) Current must change smoothly with no abrupt jumps

Figure: Current in inductance.

Refer to Slide 27 and 28

Capacitor Current and Voltage during Charging



(a)

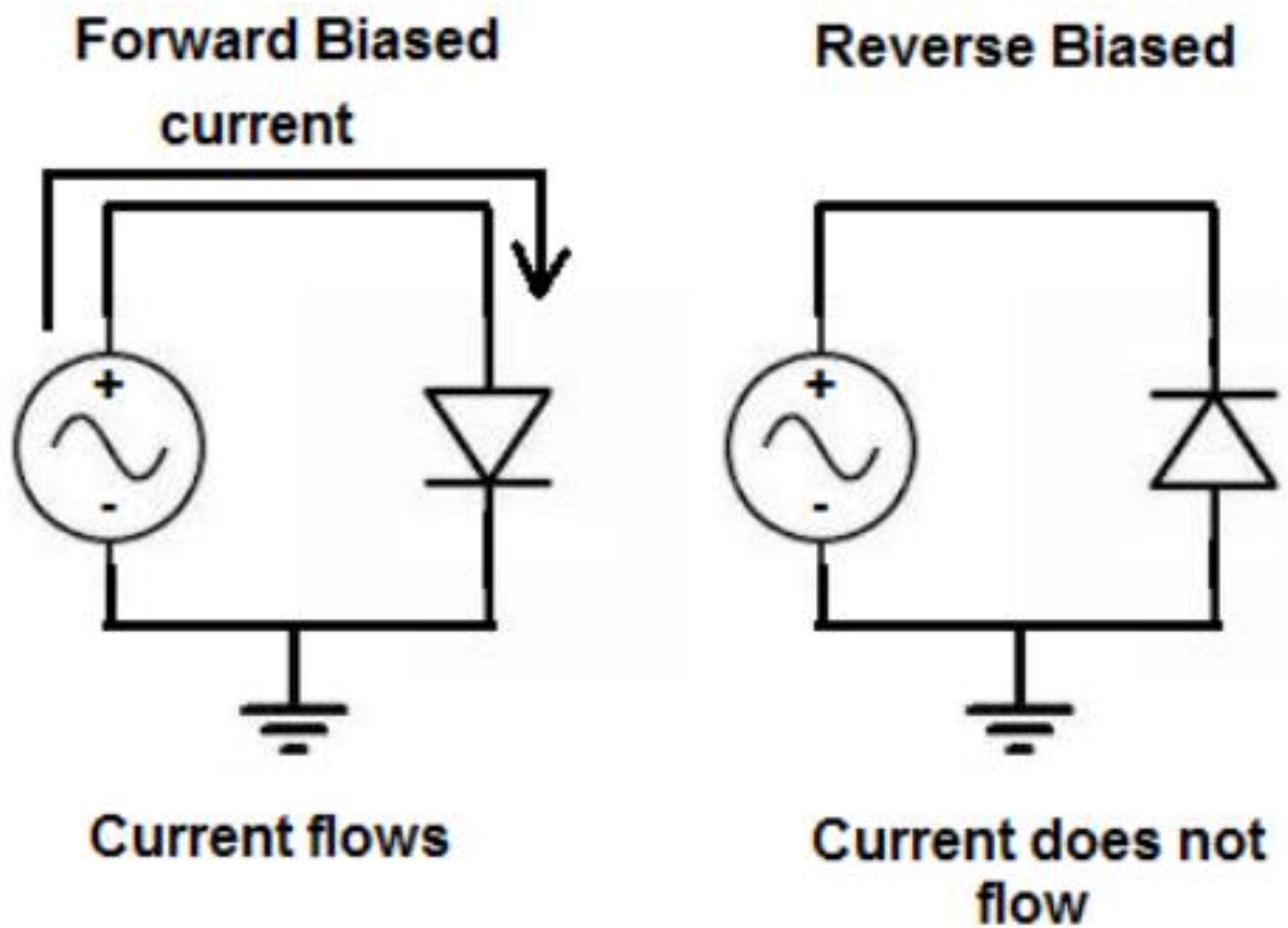
(b) Current surge during charging. Current is 0 when fully charged.

(c) Capacitor voltage. $v_C = E$ when fully charged.

Figure: The capacitor does not charge instantaneously, as a finite amount of time is required to move electrons around the circuit.

Refer to Slide 27 and 28

Forward Biased and Reverse Biased



Refer to Slide 29